1 Mapping Quantum States Directly to VSAs

One approach to model quantum states as VSAs is to represent a quantum state $|\alpha\rangle$ (suppose it is a two level system composed of n qubits) as a d dimensional hypervector. For example, the state $|00101\rangle$ would correspond to $\Phi_{00101} \in R^d$.

In order to have a representation for every possible state we would need to utilize, we would have to randomly initialize $\sum_{i=1}^{n} 2^i$ independent hypervectors. The reason we need representations for states with less than n qubits is because many quantum algorithms decompose a quantum state into the states of individual qubits within that state. This is used to perform operations and measurements on subsets of a quantum state, which due to entanglement, can affect the global quantum state. One simple example of this phenomena is quantum teleportation, where information about a quantum state can be teleported from two parties who share an pair of entangled qubits (i.e., a Bell state).

1.1 Quantum Teleportation Example

To illustrate how operations on qubit subsets affect the global state, consider quantum teleportation between Alice and Bob. The protocol begins with Alice holding an unknown qubit $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ that she wants to send to Bob. Alice and Bob also share an entangled Bell pair $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$.

The complete three-qubit system starts in the state:

$$|\psi\rangle \otimes |\Phi^+\rangle = (\alpha|0\rangle + \beta|1\rangle) \otimes \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \tag{1}$$

Alice then applies CNOT_{12} and H_1 gates (i.e., Controlled not and Hadamard gates), then performs a Bell measurement on her two qubits (the unknown state and her half of the entangled pair), as shown in Figure 1. After applying the CNOT_{12} and H_1 gates, the state becomes:

$$(\text{H}_1)(\text{CNOT}_{12})[\frac{1}{\sqrt{2}}(\alpha|000\rangle + \alpha|011\rangle + \beta|100\rangle + \beta|111\rangle)] \tag{2}$$

$$= \text{H}_1[\frac{1}{\sqrt{2}}(\alpha|000\rangle + \alpha|011\rangle + \beta|110\rangle + \beta|101\rangle)] \tag{3}$$

$$= \frac{1}{2}(\alpha|000\rangle + \alpha|100\rangle + \alpha|011\rangle + \alpha|111\rangle + \beta|010\rangle - \beta|110\rangle + \beta|001\rangle - \beta|101\rangle) \tag{4}$$

$$= \frac{1}{2}[|00\rangle(\alpha|0\rangle + \beta|1\rangle) + |01\rangle(\alpha|1\rangle + \beta|0\rangle) + |10\rangle(\alpha|0\rangle - \beta|1\rangle) + |11\rangle(\alpha|1\rangle - \beta|0\rangle)] \tag{5}$$

$$= \frac{1}{2}(|00\rangle|\psi\rangle + |01\rangle X|\psi\rangle + |10\rangle Z|\psi\rangle + ZX|11\rangle|\psi\rangle) \tag{6}$$

The above equation shows that when alice measures her two qubits, she will, with equal probability, measure either 00, 01, 10, or 11. Depending on Alice's measurement outcome, Bob's qubit becomes:

- $|\psi\rangle$ if Alice measures $|00\rangle$
- $X|\psi\rangle$ if Alice measures $|01\rangle$
- $Z|\psi\rangle$ if Alice measures $|10\rangle$
- $ZX|\psi\rangle$ if Alice measures $|11\rangle$

Alice sends her 2-bit measurement result to Bob classically, and Bob applies the corresponding Pauli correction to recover the exact state $|\psi\rangle$. This demonstrates how local operations (Alice's Bell measurement on her qubit subset) fundamentally alter the global quantum state, with Bob's qubit transforming from part of an entangled pair to carrying the teleported quantum information. Keep in mind that the coeffecients α and β are complex valued numbers, meaning they technically carry infinite bits of information, all of which have been teleported from Alice to Bob while only requiring Alice to classically send 2 bits of information to Bob, highlighing the power of the quantum teleportation algorithm.

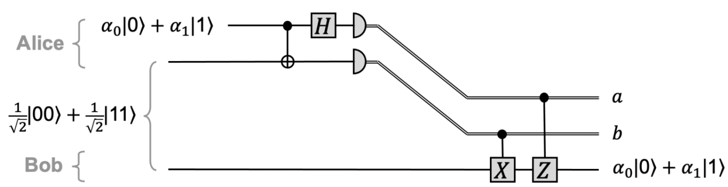

Figure 67: The teleportation protocol summarized in one circuit.

Figure 1: Taken from page 83 of `https://cleve.iqc.uwaterloo.ca/resources/QIC-710-F25/Qic710LectureNotes2025V6.pdf`

2 Structured VSA Approach: Slot-Filler Representations

2.1 Single-Qubit Representation

A more structured approach uses slot-filler representations, where we encode both the qubit position (slot) and its state (filler). For a single qubit in state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, we represent this as:

$$\Phi_\psi = \alpha(\Phi_{qubit_0} \otimes \Phi_{state_0}) + \beta(\Phi_{qubit_0} \otimes \Phi_{state_1}) \tag{7}$$

where $\Phi_{qubit_i} \in \mathbb{R}^d$ represents the i-th qubit slot and $\Phi_{state_j} \in \mathbb{R}^d$ represents the state $|j\rangle$. The binding operation $\otimes$ (often element-wise multiplication in VSAs) associates slots with their values.

This representation captures single-qubit superpositions and allows us to:

- Extract state information by unbinding with the qubit identifier

- Perform single-qubit operations by manipulating the bound vectors
- Maintain the amplitude information through the coefficients α and β

2.2 Two-Qubit Systems: Where Slot-Filler Breaks Down

Consider extending this to two-qubit Bell states. For $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$:

$$\begin{aligned}\Phi_{\Phi^+} = \frac{1}{\sqrt{2}}[(&\Phi_{qubit_0} \otimes \Phi_{state_0} + \Phi_{qubit_1} \otimes \Phi_{state_0}) \\ &+ (\Phi_{qubit_0} \otimes \Phi_{state_1} + \Phi_{qubit_1} \otimes \Phi_{state_1})]\end{aligned} \tag{8}$$

And for $|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$:

$$\begin{aligned}\Phi_{\Psi^+} = \frac{1}{\sqrt{2}}[(&\Phi_{qubit_0} \otimes \Phi_{state_0} + \Phi_{qubit_1} \otimes \Phi_{state_1}) \\ &+ (\Phi_{qubit_0} \otimes \Phi_{state_1} + \Phi_{qubit_1} \otimes \Phi_{state_0})]\end{aligned} \tag{9}$$

Since addition in VSAs is commutative, both expressions can be rewritten as:

$$\Phi_{qubit_0} \otimes \Phi_{state_0} + \Phi_{qubit_0} \otimes \Phi_{state_1} + \Phi_{qubit_1} \otimes \Phi_{state_0} + \Phi_{qubit_1} \otimes \Phi_{state_1} \tag{10}$$

This yields:

$$\Phi_{\Phi^+} = \Phi_{\Psi^+} \tag{11}$$

This result is problematic for quantum computing. The states $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ and $|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$ are orthogonal Bell states with distinct properties:

- They yield different measurement outcomes
- They transform differently under local operations
- They are used for different quantum protocols

The slot-filler approach fails to preserve the crucial correlation information that distinguishes these entangled states.

2.3 Adding Term Identifiers to Preserve Ordering

To resolve this ambiguity, we introduce term identifiers Φ_{term_i} that uniquely label each term in a superposition. For $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$:

$$\begin{aligned}\Phi_{\Phi^+} = \frac{1}{\sqrt{2}}[&\Phi_{term_1} \otimes (\Phi_{qubit_0} \otimes \Phi_{state_0} + \Phi_{qubit_1} \otimes \Phi_{state_0}) + \Phi_{term_2} \\ &\otimes (\Phi_{qubit_0} \otimes \Phi_{state_1} + \Phi_{qubit_1} \otimes \Phi_{state_1})]\end{aligned} \tag{12}$$

And for $|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$:

$$\Phi_{\Psi^+} = \frac{1}{\sqrt{2}}[\Phi_{term_1} \otimes (\Phi_{qubit_0} \otimes \Phi_{state_0} + \Phi_{qubit_1} \otimes \Phi_{state_1}) + \Phi_{term_2} \otimes (\Phi_{qubit_0} \otimes \Phi_{state_1} + \Phi_{qubit_1} \otimes \Phi_{state_0})] \tag{13}$$

These representations are now distinct. The term identifiers preserve which qubits are correlated, maintaining the essential difference between $|00\rangle + |11\rangle$ and $|01\rangle + |10\rangle$.

2.4 Modeling Quantum Gates in VSA

To implement quantum gates in this framework, we model gates based on their action on computational basis states. For the Hadamard gate, which transforms:

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \tag{14}$$

$$H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \tag{15}$$

We represent the Hadamard gate as a VSA transformation:

$$\Phi_H = \Phi_{state_0}^{-1} \otimes \frac{1}{\sqrt{2}}(\Phi_{state_0} + \Phi_{state_1}) + \Phi_{state_1}^{-1} \otimes \frac{1}{\sqrt{2}}(\Phi_{state_0} - \Phi_{state_1}) \tag{16}$$

where $\Phi_{state_i}^{-1}$ represents the inverse (or unbinding) operation for state i.

2.4.1 Application to Single-Qubit Basis States

For the basis state $|0\rangle$, represented as $\Phi_{qubit_0} \otimes \Phi_{state_0}$, applying the Hadamard gate yields:

$$\Phi_{H(|0\rangle)} = \Phi_H \otimes (\Phi_{qubit_0} \otimes \Phi_{state_0}) \tag{17}$$

$$= [\Phi_{state_0}^{-1} \otimes \frac{1}{\sqrt{2}}(\Phi_{state_0} + \Phi_{state_1})] \otimes (\Phi_{qubit_0} \otimes \Phi_{state_0}) \tag{18}$$

The $\Phi_{state_0}^{-1}$ component unbinds with Φ_{state_0}, activating only the first term of Φ_H:

$$\Phi_{H(|0\rangle)} = \Phi_{qubit_0} \otimes \frac{1}{\sqrt{2}}(\Phi_{state_0} + \Phi_{state_1}) \tag{19}$$

Similarly, for $|1\rangle$ represented as $\Phi_{qubit_0} \otimes \Phi_{state_1}$:

$$\Phi_{H(|1\rangle)} = \Phi_{qubit_0} \otimes \frac{1}{\sqrt{2}}(\Phi_{state_0} - \Phi_{state_1}) \tag{20}$$

This demonstrates that the VSA gate representation correctly reproduces the Hadamard transformation for computational basis states.

2.5 Gate Application to Superposition States

Consider applying the Hadamard gate to a state already in superposition, such as $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$. In our VSA representation with term identifiers:

$$\Phi_{|+\rangle} = \frac{1}{\sqrt{2}}[\Phi_{term_1} \otimes \Phi_{qubit_0} \otimes \Phi_{state_0} + \Phi_{term_2} \otimes \Phi_{qubit_0} \otimes \Phi_{state_1}] \quad (21)$$

Applying the Hadamard gate Φ_H and distributing:

$$\begin{aligned}\Phi_{H(|+\rangle)} = \frac{1}{2}[&\Phi_{term_1} \otimes \Phi_{qubit_0} \otimes \Phi_{state_0} + \Phi_{term_1} \otimes \Phi_{qubit_0} \otimes \Phi_{state_1} \\ &+ \Phi_{term_2} \otimes \Phi_{qubit_0} \otimes \Phi_{state_0} - \Phi_{term_2} \otimes \Phi_{qubit_0} \otimes \Phi_{state_1}]\end{aligned} \quad (22)$$

The quantum mechanical result should be $H|+\rangle = |0\rangle$, requiring the $|0\rangle$ components to constructively interfere and the $|1\rangle$ components to destructively interfere. However, $\Phi_{term_1} \otimes \Phi_{qubit_0} \otimes \Phi_{state_0}$ and $\Phi_{term_2} \otimes \Phi_{qubit_0} \otimes \Phi_{state_0}$ are treated as distinct vectors due to their different term identifiers. They cannot interfere because the VSA has no mechanism to recognize that these represent the same quantum state $|0\rangle$ that should be combined.

This reveals that while term identifiers solve the ordering problem (distinguishing $|00\rangle + |11\rangle$ from $|01\rangle + |10\rangle$), they prevent the interference effects essential to quantum computation.

3 Non-Commutative VSA Binding for Quantum States

3.1 Representation Without Term Identifiers

Using a non-commutative binding operation $\star$ where $\Phi_a \star \Phi_b \neq \Phi_b \star \Phi_a$, we can represent quantum states without term identifiers. For two-qubit states:

- $|00\rangle \to \Phi_{qubit_0} \star \Phi_{state_0} + \Phi_{qubit_1} \star \Phi_{state_0}$
- $|01\rangle \to \Phi_{qubit_0} \star \Phi_{state_0} + \Phi_{qubit_1} \star \Phi_{state_1}$
- $|10\rangle \to \Phi_{qubit_0} \star \Phi_{state_1} + \Phi_{qubit_1} \star \Phi_{state_0}$
- $|11\rangle \to \Phi_{qubit_0} \star \Phi_{state_1} + \Phi_{qubit_1} \star \Phi_{state_1}$

The Bell states now remain distinct:

$$\begin{aligned}|\Phi^+\rangle &= \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \\ &\to \frac{1}{\sqrt{2}}[(\Phi_{qubit_0} \star \Phi_{state_0} + \Phi_{qubit_1} \star \Phi_{state_0}) \\ &\qquad + (\Phi_{qubit_0} \star \Phi_{state_1} + \Phi_{qubit_1} \star \Phi_{state_1})]\end{aligned} \quad (23)$$

$$
\begin{aligned}
|\Psi^+\rangle &= \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle) \\
&\to \frac{1}{\sqrt{2}}[(\Phi_{qubit_0} \star \Phi_{state_0} + \Phi_{qubit_1} \star \Phi_{state_1}) \\
&\qquad\qquad + (\Phi_{qubit_0} \star \Phi_{state_1} + \Phi_{qubit_1} \star \Phi_{state_0})]
\end{aligned} \tag{24}
$$

Unlike with commutative binding, these representations are different. The non-commutativity preserves the correlation structure: in $|\Phi^+\rangle$, both qubits have the same state in each term, while in $|\Psi^+\rangle$, the qubits have opposite states.

3.2 Hadamard Gate on Superposition States

Applying the Hadamard gate to $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ without term identifiers:

$$
\Phi_{|+\rangle} = \frac{1}{\sqrt{2}}(\Phi_{qubit_0} \star \Phi_{state_0} + \Phi_{qubit_0} \star \Phi_{state_1}), \tag{25}
$$

$$
H = \Phi_{state_0}^{-1} \otimes \frac{1}{\sqrt{2}}(\Phi_{state_0} + \Phi_{state_1}) + \Phi_{state_1}^{-1} \otimes \frac{1}{\sqrt{2}}(\Phi_{state_0} - \Phi_{state_1}) \tag{26}
$$

Applying H and distributing:

$$
\begin{aligned}
H\Phi_{|+\rangle} &= \frac{1}{\sqrt{2}}[\frac{1}{\sqrt{2}}(\Phi_{qubit_0} \star \Phi_{state_0} + \Phi_{qubit_0} \star \Phi_{state_1}) \\
&\qquad\qquad + \frac{1}{\sqrt{2}}(\Phi_{qubit_0} \star \Phi_{state_0} - \Phi_{qubit_0} \star \Phi_{state_1})] \\
&= \Phi_{qubit_0} \star \Phi_{state_0}
\end{aligned} \tag{27}
$$

The interference operates correctly because terms with the same qubit-state binding naturally combine through vector addition. The non-commutative binding provides the ordering information needed for entanglement without preventing interference.

3.3 Analysis

The non-commutative binding resolves both fundamental issues:

1. **Preserves entanglement structure**: Different Bell states such as $|00\rangle + |11\rangle$ and $|01\rangle + |10\rangle$ map to distinct VSA representations

2. **Allows interference**: Gates can operate on superpositions correctly since like terms naturally combine

The non-commutativity of $\star$ provides the ordering information that was previously encoded in term identifiers, but without preventing the interference effects essential to quantum computation.

4 Qubit-Specific VSA Representation with HRR Binding

4.1 State Representation

We propose a VSA representation where each qubit i has two dedicated hypervectors: $\Phi_{q_i^0} \in \mathbb{R}^d$ representing $|0\rangle$ at position i, and $\Phi_{q_i^1} \in \mathbb{R}^d$ representing $|1\rangle$ at position i. These hypervectors are randomly initialized and remain fixed throughout computation.

For an n-qubit quantum state, we construct the VSA representation by binding the appropriate qubit-state vectors using the HRR binding operation $\circledast$. For a computational basis state $|b_1 b_2 ... b_n\rangle$ where $b_i \in \{0, 1\}$:

$$\Phi_{|b_1 b_2 ... b_n\rangle} = \Phi_{q_1^{b_1}} \circledast \Phi_{q_2^{b_2}} \circledast ... \circledast \Phi_{q_n^{b_n}} \tag{28}$$

For a general superposition state $|\psi\rangle = \sum_i \alpha_i |i\rangle$, the VSA representation becomes:

$$\Phi_{|\psi\rangle} = \sum_i \alpha_i \Phi_{|i\rangle} \tag{29}$$

For example, the two-qubit uniform superposition state:

$$|\psi\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle) \tag{30}$$

is represented as:

$$\Phi_{|\psi\rangle} = \frac{1}{2}(\Phi_{q_0^0} \circledast \Phi_{q_1^0} + \Phi_{q_0^0} \circledast \Phi_{q_1^1} + \Phi_{q_0^1} \circledast \Phi_{q_1^0} + \Phi_{q_0^1} \circledast \Phi_{q_1^1}) \tag{31}$$

This representation maintains distinctness between different quantum states. For instance, the Bell states:

$$\begin{aligned} |\Phi^+\rangle &= \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \\ &\rightarrow \frac{1}{\sqrt{2}}(\Phi_{q_0^0} \circledast \Phi_{q_1^0} + \Phi_{q_0^1} \circledast \Phi_{q_1^1}) \end{aligned} \tag{32}$$

$$\begin{aligned} |\Psi^+\rangle &= \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle) \\ &\rightarrow \frac{1}{\sqrt{2}}(\Phi_{q_0^0} \circledast \Phi_{q_1^1} + \Phi_{q_0^1} \circledast \Phi_{q_1^0}) \end{aligned} \tag{33}$$

map to distinct VSA representations despite the commutativity of HRR binding.

4.2 Measurement

To measure qubit k in the computational basis, we compute the similarity between the state vector and the basis vectors for that qubit. Given a state $\Phi_{|\psi\rangle}$, the measurement process involves:

1. Computing similarity scores:

$$s_0 = \langle \Phi_{|\psi\rangle}, \Phi_{q_k^0} \rangle \tag{34}$$

$$s_1 = \langle \Phi_{|\psi\rangle}, \Phi_{q_k^1} \rangle \tag{35}$$

2. Determining measurement probabilities:

$$P(0) = \frac{|s_0|^2}{|s_0|^2 + |s_1|^2} \tag{36}$$

$$P(1) = \frac{|s_1|^2}{|s_0|^2 + |s_1|^2} \tag{37}$$

3. Collapsing the state based on measurement outcome $m \in \{0, 1\}$:

$$\Phi_{|\psi'\rangle} = \frac{1}{\sqrt{P(m)}} \text{Extract}(\Phi_{|\psi\rangle}, \Phi_{q_k^m}) \tag{38}$$

where $\text{Extract}(\Phi, \Phi_{q_k^m})$ retains only components of Φ that contain $\Phi_{q_k^m}$ in their binding structure.

4.3 Partial Measurements

The measurement operation described in the previous section applies only to complete measurements of all qubits. Partial measurements, where we measure a subset of qubits while leaving others in superposition, require a more sophisticated approach.

4.3.1 Measurement Probability Calculation

Consider an n-qubit state $\Phi_{|\psi\rangle}$ where we wish to measure qubit k. To determine the measurement probabilities, we must compute the overlap with all computational basis states that have the same value for qubit k.

For measuring qubit k to obtain outcome $m \in \{0, 1\}$, we compute:

$$s_m = \sum_{b_1, \ldots, b_{k-1}, b_{k+1}, \ldots, b_n \in \{0,1\}} |\langle \Phi_{|\psi\rangle}, \Phi_{|b_1 \ldots b_{k-1} m b_{k+1} \ldots b_n\rangle} \rangle|^2 \tag{39}$$

The measurement probability is then:

$$P(m) = s_m \tag{40}$$

Note that if the state is properly normalized, $s_0 + s_1 = 1$.

4.3.2 Example: Two-Qubit Partial Measurement

Consider the normalized state:

$$|\psi\rangle = \frac{1}{2}(\sqrt{2}|00\rangle + |01\rangle + |11\rangle) \tag{41}$$

represented as:

$$\Phi_{|\psi\rangle} = \frac{1}{2}(\sqrt{2}\Phi_{q_0^0} \circledast \Phi_{q_1^0} + \Phi_{q_0^0} \circledast \Phi_{q_1^1} + \Phi_{q_0^1} \circledast \Phi_{q_1^1}) \tag{42}$$

To measure the first qubit, we compute:

$$\begin{aligned} s_0 &= |\langle \Phi_{|\psi\rangle}, \Phi_{|00\rangle}\rangle|^2 + |\langle \Phi_{|\psi\rangle}, \Phi_{|01\rangle}\rangle|^2 & (43)\\ &= |\langle \Phi_{|\psi\rangle}, \Phi_{q_0^0} \circledast \Phi_{q_1^0}\rangle|^2 + |\langle \Phi_{|\psi\rangle}, \Phi_{q_0^0} \circledast \Phi_{q_1^1}\rangle|^2 & (44) \end{aligned}$$

$$\begin{aligned} s_1 &= |\langle \Phi_{|\psi\rangle}, \Phi_{|10\rangle}\rangle|^2 + |\langle \Phi_{|\psi\rangle}, \Phi_{|11\rangle}\rangle|^2 & (45)\\ &= |\langle \Phi_{|\psi\rangle}, \Phi_{q_0^1} \circledast \Phi_{q_1^0}\rangle|^2 + |\langle \Phi_{|\psi\rangle}, \Phi_{q_0^1} \circledast \Phi_{q_1^1}\rangle|^2 & (46) \end{aligned}$$

Due to the orthogonality of the basis states in VSA:

$$s_0 = \left|\frac{\sqrt{2}}{2}\right|^2 + \left|\frac{1}{2}\right|^2 = \frac{2}{4} + \frac{1}{4} = \frac{3}{4} \tag{47}$$

$$s_1 = |0|^2 + \left|\frac{1}{2}\right|^2 = 0 + \frac{1}{4} = \frac{1}{4} \tag{48}$$

Therefore, $P(0) = \frac{3}{4}$ and $P(1) = \frac{1}{4}$.

4.3.3 State Collapse After Measurement

Once a measurement outcome m is obtained (sampled from the probability distribution), the state must be collapsed accordingly. The collapse operation is performed by unbinding with the measured qubit value:

$$\Phi_{|\psi_{collapsed}\rangle} = \frac{1}{\sqrt{P(m)}} \cdot \Phi_{|\psi\rangle} \circledast (\Phi_{q_k^m})^{-1} \tag{49}$$

This unbinding operation effectively:

- Preserves terms compatible with the measurement outcome
- Zeros out (approximately) terms incompatible with the measurement
- Reduces the dimensionality of the quantum state by one qubit

For our example, if we measure 0 on the first qubit:

$$\Phi_{|\psi_{collapsed}\rangle} = \frac{1}{\sqrt{3/4}} \cdot \Phi_{|\psi\rangle} \circledast (\Phi_{q_0^0})^{-1} \tag{50}$$

$$\approx \frac{1}{\sqrt{3}}(\sqrt{2}\Phi_{q_1^0} + \Phi_{q_1^1}) \tag{51}$$

The resulting state represents the second qubit in the superposition $\frac{1}{\sqrt{3}}(\sqrt{2}|0\rangle + |1\rangle)$, which is the correct post-measurement state.

4.3.4 Scaling to Larger Systems

For an n-qubit system where we measure k qubits, the number of basis states to compute overlaps with is 2^k. While this grows exponentially with the number of measured qubits, it remains tractable for small partial measurements and is significantly more efficient than maintaining the full 2^n-dimensional state vector.

The unbinding operation for multiple qubit measurements follows naturally:

$$\Phi_{|\psi_{collapsed}\rangle} = \frac{1}{\sqrt{P(m_1, m_2, ..., m_k)}} \cdot \Phi_{|\psi\rangle} \circledast (\Phi_{q_{i_1}^{m_1}})^{-1} \circledast (\Phi_{q_{i_2}^{m_2}})^{-1} \circledast ... \circledast (\Phi_{q_{i_k}^{m_k}})^{-1} \tag{52}$$

where $i_1, i_2, ..., i_k$ are the indices of the measured qubits and $m_1, m_2, ..., m_k$ are their respective measurement outcomes.

The key advantage of this approach is that the VSA representation maintains constant vector dimension d regardless of the number of qubits. The tradeoff is that as the number of qubits increases, the accumulated noise from repeated binding operations reduces the fidelity of the representation. However, for quantum algorithms that require partial measurements on small subsets of qubits, this approach provides a practical compromise between memory efficiency and computational accuracy.

This partial measurement scheme preserves the essential quantum mechanical properties: measurement probabilities follow the Born rule, state collapse occurs appropriately, and unmeasured qubits remain in superposition. The approach fails only when the accumulated noise overwhelms the signal, which occurs at a system-size threshold dependent on the hypervector dimension d and the specific quantum algorithm being simulated.

4.3.5 Hadamard Gate Example

The Hadamard gate transforms:

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \tag{53}$$

$$H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) \tag{54}$$

In VSA representation, the Hadamard gate on qubit k is:

$$\Phi_{H_k} = \Phi_{q_k^0}^{-1} \circledast \frac{1}{\sqrt{2}}(\Phi_{q_k^0} + \Phi_{q_k^1}) + \Phi_{q_k^1}^{-1} \circledast \frac{1}{\sqrt{2}}(\Phi_{q_k^0} - \Phi_{q_k^1}) \tag{55}$$

Applying this to a single-qubit state $|0\rangle$ represented as $\Phi_{q_k^0}$:

$$\Phi_{H_k} \circledast \Phi_{q_k^0} = \frac{1}{\sqrt{2}}(\Phi_{q_k^0} + \Phi_{q_k^1}) \tag{56}$$

For multi-qubit states, the gate application follows the same principle. Consider applying H_0 to the state $|00\rangle = \Phi_{q_0^0} \circledast \Phi_{q_1^0}$:

$$\Phi_{H_0(|00\rangle)} = \Phi_{H_0} \circledast (\Phi_{q_0^0} \circledast \Phi_{q_1^0}) \tag{57}$$

$$= \frac{1}{\sqrt{2}}(\Phi_{q_0^0} + \Phi_{q_0^1}) \circledast \Phi_{q_1^0} \tag{58}$$

$$= \frac{1}{\sqrt{2}}(\Phi_{q_0^0} \circledast \Phi_{q_1^0} + \Phi_{q_0^1} \circledast \Phi_{q_1^0}) \tag{59}$$

4.3.6 Two-Qubit Gates

For two-qubit gates such as CNOT, the VSA representation extends to include transformations on bound pairs. The CNOT gate with control qubit c and target qubit t:

$$\begin{aligned}\Phi_{CNOT_{c,t}} = {} & (\Phi_{q_c^0} \circledast \Phi_{q_t^0})^{-1} \circledast (\Phi_{q_c^0} \circledast \Phi_{q_t^0}) + (\Phi_{q_c^0} \circledast \Phi_{q_t^1})^{-1} \circledast (\Phi_{q_c^0} \circledast \Phi_{q_t^1}) \\ & + (\Phi_{q_c^1} \circledast \Phi_{q_t^0})^{-1} \circledast (\Phi_{q_c^1} \circledast \Phi_{q_t^1}) + (\Phi_{q_c^1} \circledast \Phi_{q_t^1})^{-1} \circledast (\Phi_{q_c^1} \circledast \Phi_{q_t^0})\end{aligned} \tag{60}$$

In general, since any gate, regardless of the number of qubits it operates on, can be represented as a unitary matrix, it is possible interpret the gate as transforming basis states according to the column corresponding to that basis state. In this way, we can create a VSA mapping for any quantum gate.

4.4 Complexity and Noise Considerations

This approach requires only $2n$ hypervectors for an n-qubit system, providing linear scaling rather than exponential. The tradeoff is accumulated noise through repeated binding operations, particularly for systems with many qubits. The noise increases with circuit depth and qubit count, but the memory requirement remains constant at $\mathcal{O}(nd)$ where d is the hypervector dimension.